

Comparing Logistic Regression and Random Forest for Binary Classification

1 Introduction

This study compares a regularized linear classifier with a nonlinear tree ensemble on the supplied tabular dataset. The objective is a reproducible workflow with justified cleaning, preprocessing and evaluation decisions. The positive class is "yes". Model settings are selected using training-only cross-validation; the supplied test partition is reserved for the final comparison.

2 Data exploration and cleaning

The files contain 31,112 training and 13,334 test records, with 15 predictors: seven numerical and eight categorical. Three training records and one test record have missing labels; these are excluded because supervised targets cannot be inferred reliably. The remaining training set contains 31,109 records (7,464 yes; 23,645 no), and the evaluation set contains 13,333 records (3,168 yes; 10,165 no). The positive proportions are 24.0% and 23.8%, respectively, indicating moderate class imbalance.

The original training file has 47 missing predictor cells, fewer than 0.02% of predictor entries. Numerical gaps are filled with training medians, which are less sensitive to extreme values than means; categorical gaps use training modes, a simple choice given the very small missing fraction. No exact duplicate rows occur within either file; 0 test feature rows exactly match training feature rows. These checks cannot rule out repeated entities because no entity identifier is supplied.

Training summaries show different feature scales: `performance_score` ranges from 0 to 1,010, while `stability_index` ranges from 0 to 4.625. For `load_ratio`, the median is 23.689, the middle 50% spans 19.272-27.657, and the maximum is 143.717. This large upper tail supports median imputation and illustrates the distribution checks taught in Lecture 3 [3].

Training-set 1.5-IQR fences flag 5,347 `activity_duration` values and 2,526 `load_ratio` values. These are statistical tail observations, not verified errors. Lecture 2 [3] emphasizes task-dependent cleaning: without a data dictionary or invalid-value rules, deleting extremes could remove useful class patterns. They are retained, while acknowledging that logistic regression may be sensitive to them; clipping requires a training-only sensitivity study.

3 Features and experimental procedure

All 15 predictors are retained. One-hot encoding represents nominal categories without artificial order; unknown categories map to zeros. Numerical features are standardized for logistic regression so regularization is comparable across scales; the forest needs no scaling. L2 regularization stabilizes correlated coefficients; no PCA is applied because preserving original feature meaning is preferable here. No interactions, resampling or label-derived features are added. Imputation, encoding and scaling are refitted inside each validation fold, preventing preprocessing leakage.

Both models use shuffled, stratified three-fold cross-validation (seed 42), preserving class proportions with modest computation. Following Lecture 4's emphasis on precision-recall evaluation under imbalance [3], grid search selects mean average precision (AP). AP is recall-weighted precision across thresholds, not trapezoidal PR-AUC. Each selected pipeline is refitted on all labelled training data, then evaluated once on the test set. The threshold is fixed at 0.50. Folds and seed are experimental settings, distinct from model hyperparameters (Lecture 5 [3]).

Logistic regression [1] uses L2 regularization, the LBFGS solver, C in $\{0.1, 1, 10\}$, tolerance 0.0001 and a maximum of 2,000 iterations. Random forest [2] uses 200 bootstrap trees, Gini splits, sqrt feature subsampling, maximum depth in $\{12, \text{unrestricted}\}$, and minimum leaf size in $\{2, 10\}$. Trees stop at the selected depth or when no admissible split remains; training stops after 200 trees. Both models use default unweighted classes. This preserves observed class proportions but may sacrifice positive recall; no unequal error costs were supplied.

4 Results

The selected logistic regression uses $C = 10$; the selected forest uses maximum depth = 12 and minimum leaf size = 2. Their cross-validation AP values are 0.7085 (fold SD 0.0108) and 0.7130 (fold SD 0.0145), respectively. These fold standard deviations describe split variability, not confidence intervals.

Table 1. Held-out test performance; precision, recall and F1 refer to the yes class.

Metric	Always no	Logistic regression	Random forest
Accuracy	0.7624	0.8396	0.8390
Balanced accuracy	0.5000	0.7473	0.7311
Precision	0.0000	0.6986	0.7211
Recall	0.0000	0.5713	0.5256
F1	0.0000	0.6286	0.6080
ROC-AUC	0.5000	0.8888	0.8895
Average precision	0.2376	0.7025	0.7067

The always-no baseline has zero positive recall and F1 despite 76.2% accuracy, illustrating why accuracy alone is insufficient. Its precision is set to zero by convention because it predicts no positives. Balanced accuracy averages sensitivity and specificity; ROC-AUC and AP assess ranking without fixing a classification threshold. The baseline AP of 0.2376 equals test positive prevalence.

Table 2. Test confusion counts at threshold 0.50.

Model	True no	False yes	Missed yes	True yes
Logistic regression	9384	781	1358	1810
Random forest	9521	644	1503	1665

5 Discussion

The forest improves test AP by 0.0043, but logistic regression has higher F1 and recall at 0.50. The forest combines bootstrap trees and random feature subsets, as covered in Lecture 5 [3]. With leaf size 2, restricting depth to 12 raises validation AP from 0.7072 to 0.7130 compared with unrestricted trees. This supports complexity control (Lecture 4 [3]), although it does not prove overfitting without training-score analysis.

Relative to logistic regression, the forest avoids 137 false positives but misses 145 additional positives (Table 2). Model choice therefore depends on error costs. If missed positives are expensive, a lower threshold or class weighting should be evaluated on training-only validation data, reporting the precision cost. Calibration has not been assessed, so scores should not automatically be interpreted as reliable individual risks.

The small AP difference does not establish statistical significance. Both models use the same test records, so an independent-test-set comparison would be inappropriate; paired uncertainty analysis would be needed. The limited grid and one split constrain generalization. Without domain, entity or time metadata, independence cannot be verified and semantic target leakage cannot be excluded. Next steps are repeated or grouped validation where appropriate, preprocessing sensitivity checks and a fresh external test. No causal interpretation is warranted.

6 Conclusion and reproducibility

The workflow demonstrates that minority-sensitive metrics reveal performance hidden by the majority baseline. Random forest provides the stronger observed AP, while the confusion counts make the operational trade-off explicit. Results are reproducible with seed 42 and scikit-learn 1.9.1; the accompanying script exports every grid result, prediction and audit count.

Source repository: <https://github.com/Davi41234/IN6227>

References

- [1] Scikit-learn. LogisticRegression. [API documentation](#).
- [2] Scikit-learn. RandomForestClassifier. [API documentation](#).
- [3] Xiang, Y. (2026). IN6227 Lectures 2-5: Data, Data Exploration, and Classification. NTU, NTULearn course slides.